

ECON 1550 International Finance

Output and the Exchange Rate in the Short Run

Behavioral equations in the goods market

$$Y = C + I + G + CA$$

$$\begin{array}{l} \text{DEMAND FOR} \\ \text{DOMESTIC} \\ \text{GOOD} \end{array} = C + I + G + CA$$

$$\begin{array}{l} \text{SUPPLY FOR} \\ \text{DOMESTIC} \\ \text{GOOD} \end{array} = Y$$

Demand for domestic goods

$$\begin{array}{l} \text{DEMAND} \\ \text{FOR DOM} \\ \text{GOODS} \end{array} = C + I + G + CA = D$$

$$\begin{array}{l} \text{DOMESTIC} \\ \text{DEMAND} \\ \text{FOR GOODS} \end{array} \equiv A = C + I + G$$

$$A + EX - IM = A + CA = D$$

Demand for domestic goods

$$C = C(Y_D^{(+1)}) \quad \text{CONSUMPTION (OF DOM. AND FOREIGN)}$$

$$Y_D \equiv Y - T \quad \text{DISPOSABLE INCOME}$$

$$\begin{array}{ll} I = \bar{I} & \text{EXOGENOUS INVESTMENT} \\ G = \bar{G} & \text{" GOV. PURCHASES} \end{array} \left. \vphantom{\begin{array}{l} I = \bar{I} \\ G = \bar{G} \end{array}} \right\} \begin{array}{l} \text{BOTH DOM} \\ \text{AND FOR.} \\ \text{GOODS} \end{array}$$

$$CA = CA(q, Y - T, Y^*)$$

$$q \equiv \frac{EP^*}{P} = \frac{\text{DOLLAR PRICE OF FOREIGN GOOD}}{\text{DOLLAR PRICE OF DOM GOOD}}$$

Demand for domestic goods

$$CA = EX - IM$$

$$EX = EX(q, Y^*)$$

(+1) (+1)

$$IM = IM(q, Y_D)$$

(-) (+1)

$q \uparrow$ VOLUME & PRICE $q \uparrow$
VOLUME DOMINATES

$q \uparrow$ REAL DEPR.

$$\frac{EP^*}{P} = q$$

$\Rightarrow$ HOME
 GOOD
 APPEARS
 CHEAPER

$$\underbrace{IM}_{\text{DOM GOOD}} = \underbrace{\frac{\text{PRICE}}{\text{DOM GOOD}}}_{\text{FOREIGN GOOD}} \times \underbrace{VOLUME}_{\text{FOREIGN GOOD}}$$

A short-run model of the goods market

Exogenous variables	
Variable	Description
E	Nominal exchange rate
I	Investment
G	Government spending
T	Taxes
P	Price level
P^*	Foreign price level
Y^*	Foreign income

Endogenous variables

Variable	Description	Equation	Type of equation
Y	Income, production	$Y = D$	Equilibrium condition
Y_D	Disposable income	$Y_D \equiv Y - T$	Identity
EX	Exports	$EX = EX(q, Y^*)$ (+) (+)	Behavioral
IM	Imports	$IM = IM(q, Y_D)$ (-) (+)	Behavioral
CA	Current account	$CA \equiv EX - IM = CA(q, Y_D, Y^*)$ (+) (-) (+)	Identity
D	Demand for domestic goods	$D \equiv C + I + G + CA$	Identity
A	Domestic demand	$A \equiv C + I + G$	Identity
C	Consumption	$C = C(Y_D)$ (+)	Behavioral
q	Real exchange rate	$q \equiv \frac{EP^*}{P}$	Identity

DD Curve

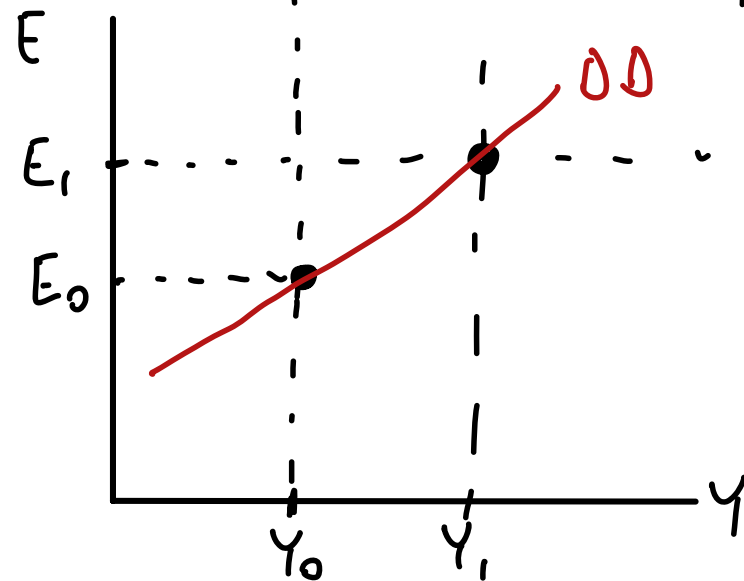
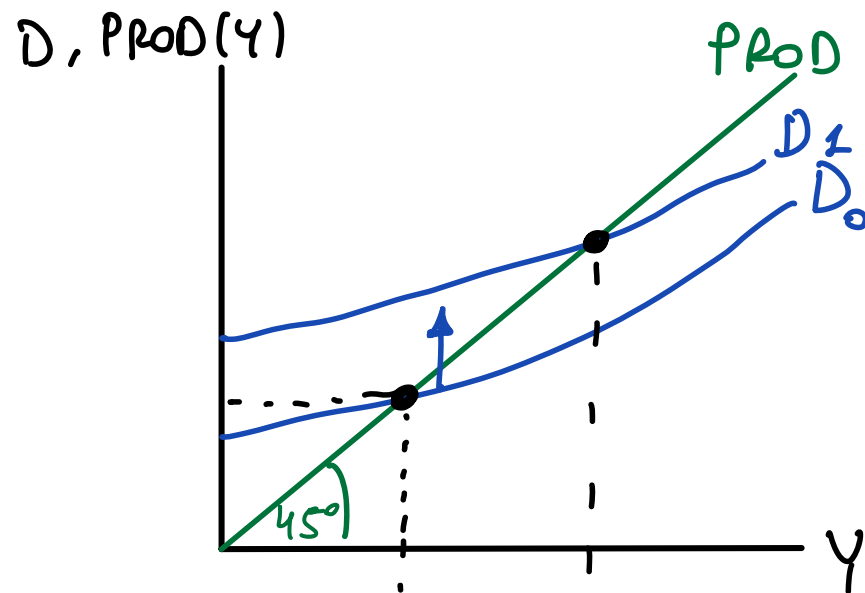
$$Y = D$$

$$Y = C + I + G + CA$$

$$Y = C_{(+)}(Y_D) + I + G + CA_{(+)}(q_{(+)} , Y_{D(-)} , Y^*_{(+)})$$

$$Y = C_{(+)}(Y_D) + I + G + CA_{(+)}(EP^*/P_{(+)} , Y_{D(-)} , Y^*_{(+)})$$

$Y \uparrow \quad C \uparrow$ BUT $CA \downarrow$. HOWEVER,
 D ALWAYS $\uparrow$



$$D = C + I + G + NX \quad \text{WHEN } Y \uparrow$$

$C \uparrow$ $NX \downarrow$ SO D ?

NOT UNCLEAR, $D \uparrow$

C : CONS. OF DOM. AND FOREIGN GOODS

$C - IM$ = DOM DEM. FOR DOM GOOD

$$NX = \underbrace{EX} - \underbrace{IM}$$

DOES NOT
CHANGE

IM GOES
UP

$$C = C_{\text{DOM GOOD}} + C_{\text{FOREIGN GOOD}}$$

IM = DOM DEMAND FOR
FOREIGN

Short-run FX and money market model

Exogenous variables

Variable	Description
R^*	Foreign interest rate
E^e	Expected exchange rate
Y	Real income
M^s	Money supply
P	Price level

Endogenous variables

Variable	Description	Equation	Type of equation
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Equilibrium condition
R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^d	Money demand	$M^d = M^s$	Equilibrium condition

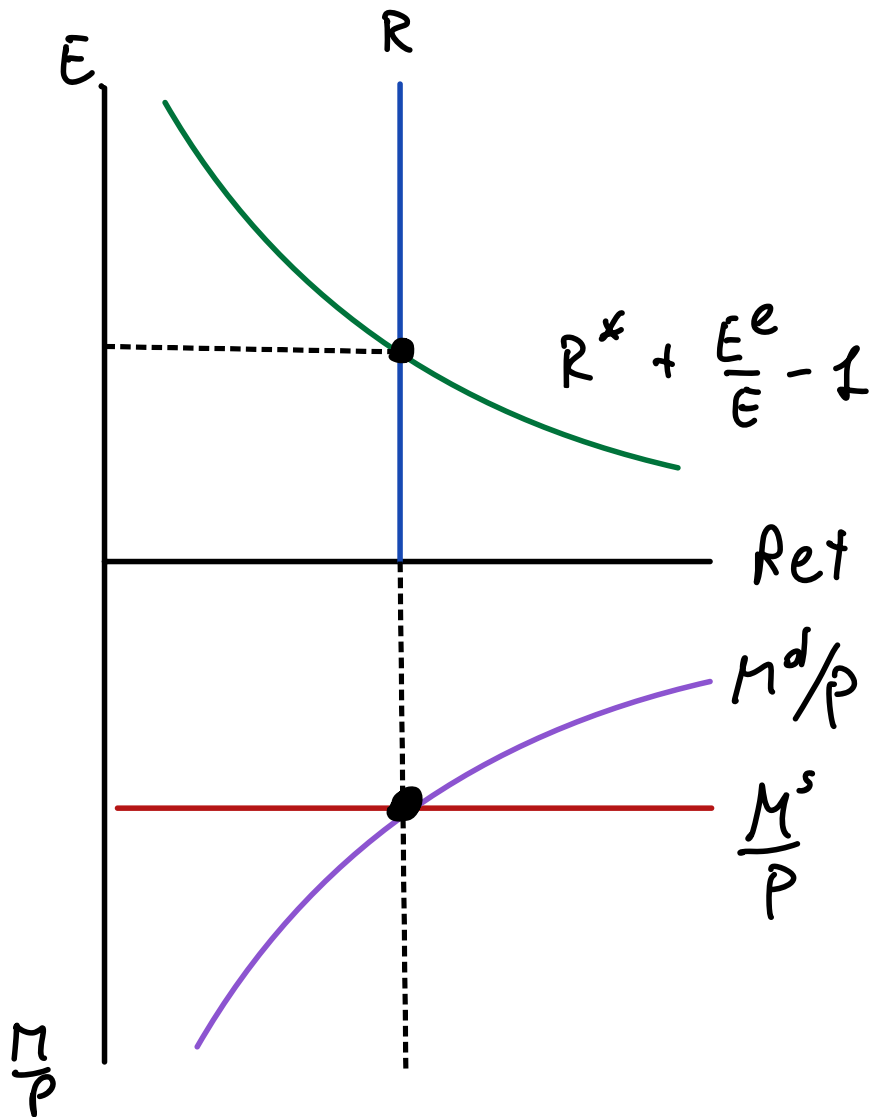
AA Curve

(UIP):
$$R = R^* + \frac{E^e}{E} - 1$$

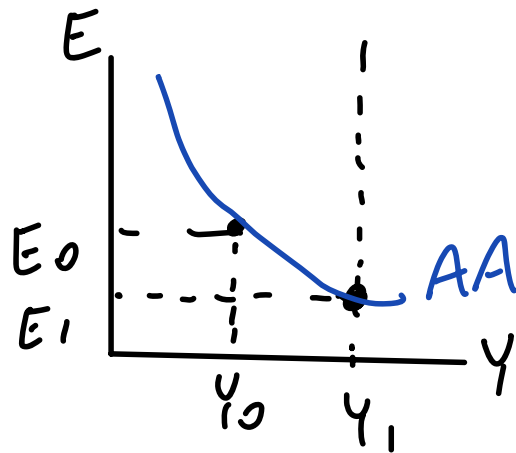
(MS) = (MD):
$$\frac{M^s}{P} = L(R, Y)$$

 (-) (+)

Y EXO } FIND E FOR
 E ENDO } GIVEN Y



AA Curve



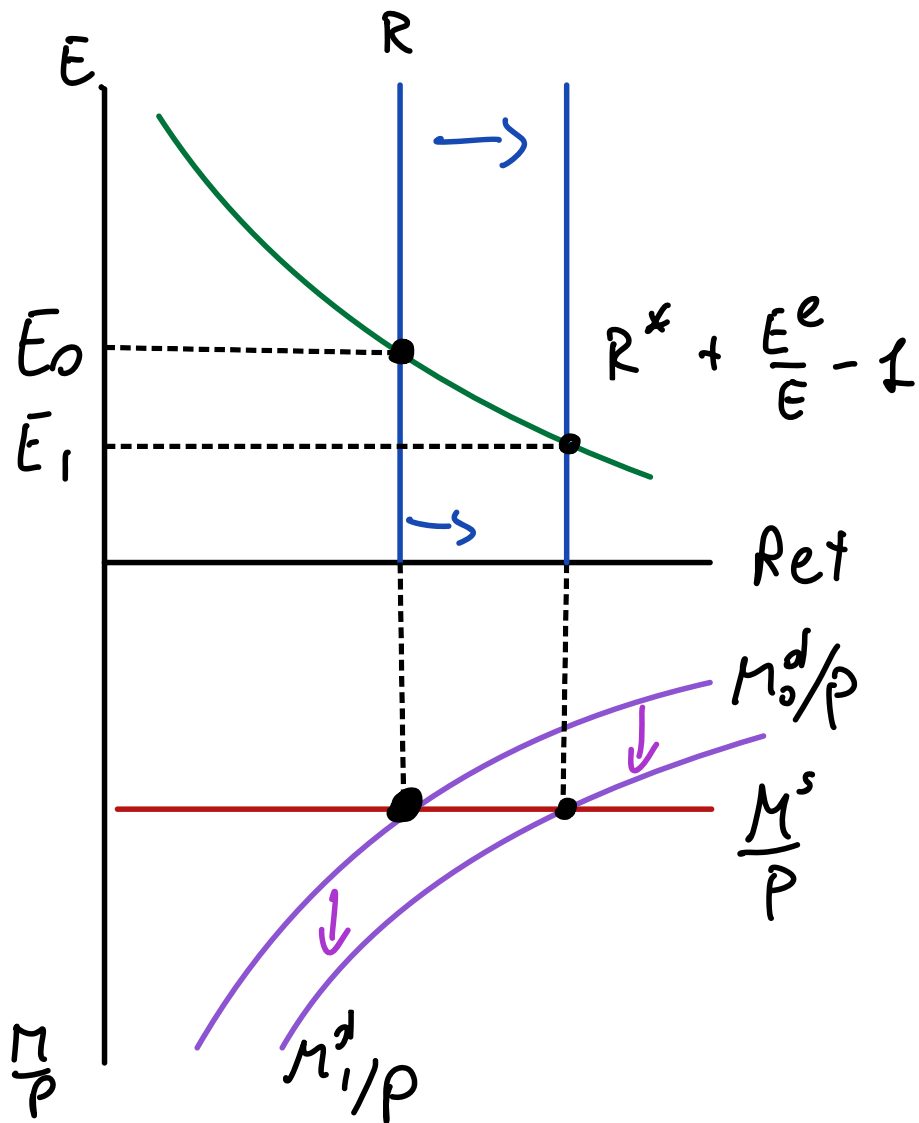
(UIP):
$$R = R^* + \frac{E^e}{E} - 1$$

(MS) = (MD):
$$\frac{M^s}{P} = L(R, Y)$$

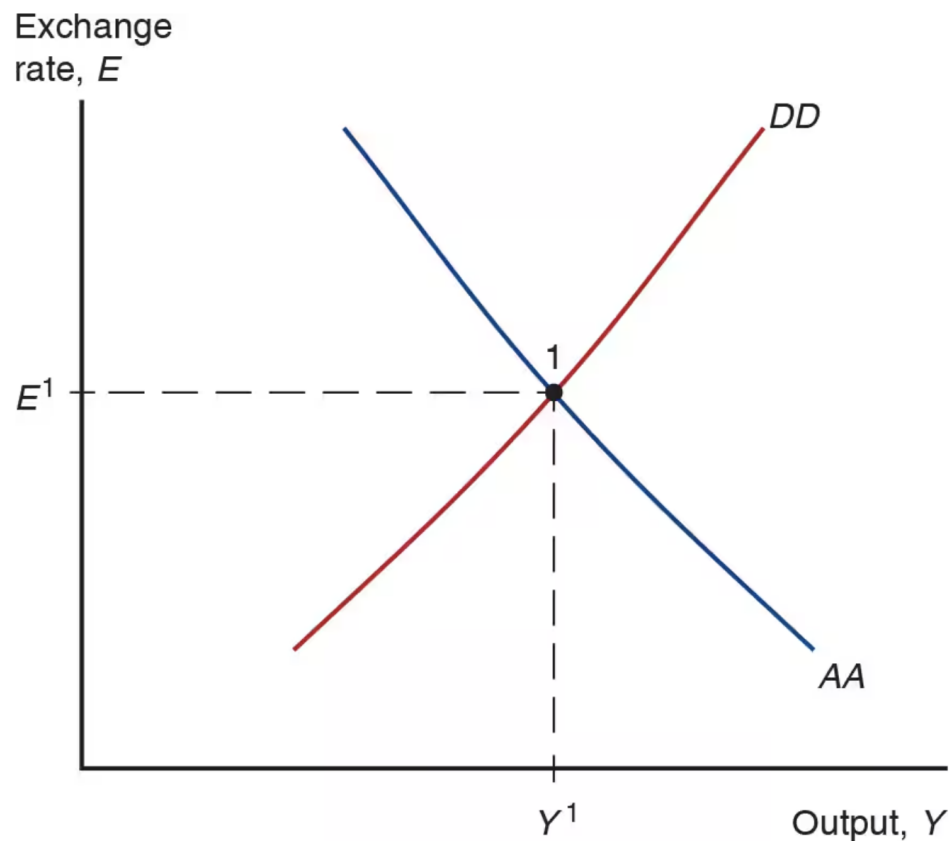
 (-) (+)

(Y_0, E_0) WITH $Y_1 > Y_0$

(Y_1, E_1) $E_1 < E_0$



Short-run Equilibrium in *AA-DD* model



GOODS

E EXO

MM + FX

Y EXO

COMBINE

Y, E ARE
BOTH

ENDO

Example: DD Schedule

$$C_{(+)}(Y_D) = 1 + 0.75Y_D$$

$$EX_{(+)}(q, Y_{(+)}^*) = 0.15 + 0.5q + 0.1Y^*$$

$$IM_{(-)}(q, Y_{(+)}^D) = 0.1 - 0.2q + 0.12Y_D$$

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

How to find the DD schedule

EQ. COND FOR GOODS:

$$Y = D$$

$$Y = C + I + G + CA$$

$$Y = 1 + 0.75(Y - T) + I + G + 0.15 + 0.5q + 0.1Y^* - 0.1 + 0.2q - 0.12(Y - T)$$

Example: DD Schedule

. CAN SOLVE FOR E
TO GET EXPLICIT
E = FUNCTION OF Y.

$$C(Y_D) = 1 + 0.75Y_D$$

(+)

$$EX(q, Y^*) = 0.15 + 0.5q + 0.1Y^*$$

(+)

$$IM(q, Y_D) = 0.1 - 0.2q + 0.12Y_D$$

(-) (+)

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

How to find the DD schedule

$$Y = 1 + 0.75(Y - T) + I + G +$$

$$+ 0.15 + 0.5q + 0.1Y^*$$

$$- 0.1 + 0.2q - 0.12(Y - T)$$

$$Y = 1 + 0.15 - 0.1 + I + G$$

$$(0.75 - 0.12)(Y - T) +$$

$$(0.5 + 0.2) \frac{EP^*}{P} + 0.1Y^*$$

Example: AA Schedule

• CAN SOLVE FOR E
TO GET:

$E = \text{FUNCTION OF } Y$

$$R = R^* + \frac{E^e - E}{E}$$

$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\begin{aligned} \frac{M^d}{P} &= L\left(\underset{(-)}{R}, \underset{(+)}{Y}\right) \\ &= 1.1 - R + 0.05Y \end{aligned}$$

How to find the AA schedule

$$\frac{M^s}{P} = \frac{M^d}{P}$$

$$\frac{M^s}{P} = 1.1 - R + 0.05Y$$

$$\frac{M^s}{P} = 1.1 - R^* - \frac{E^e}{E} + 1 + 0.05Y$$

Example: AA Schedule

• CAN SOLVE FOR E
TO GET:

$E = \text{FUNCTION OF } Y$

$$R = R^* + \frac{E^e - E}{E}$$

How to find the AA schedule

$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\frac{M^s}{P} = 1.1 - R^* - \frac{E^e}{E} + 1 + 0.5Y$$

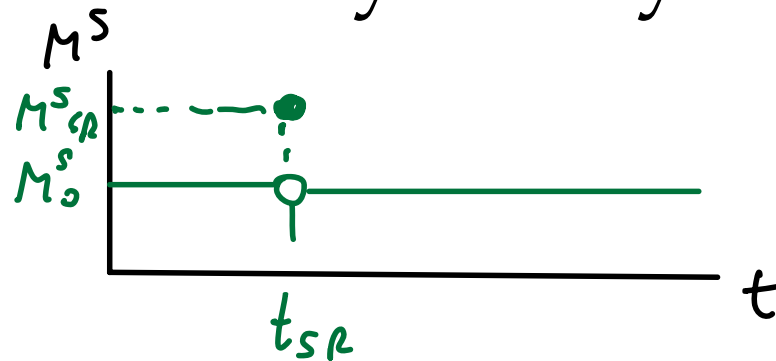
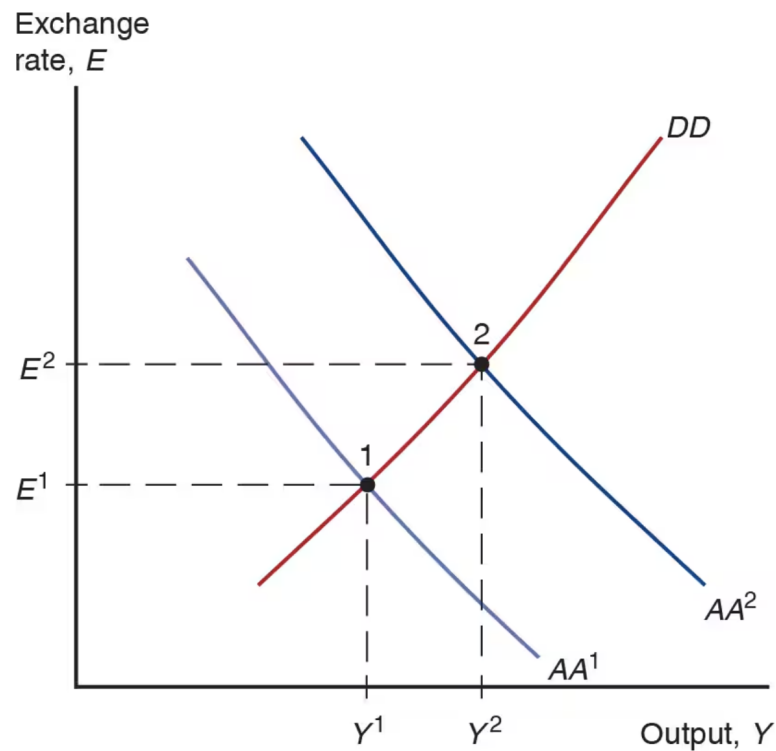
$$\begin{aligned} \frac{M^d}{P} &= L(\underset{(-)}{R}, \underset{(+)}{Y}) \\ &= 1.1 - R + 0.05Y \end{aligned}$$

SOLVE FOR E :

$$E^e/E = 1.1 - R^* + 1 - M^s/P + 0.5Y$$

$$E = \frac{E^e}{1.1 - R^* + 1 - M^s/P + 0.5Y}$$

Temporary Change in Monetary Policy



TEMPORARY SHOCKS

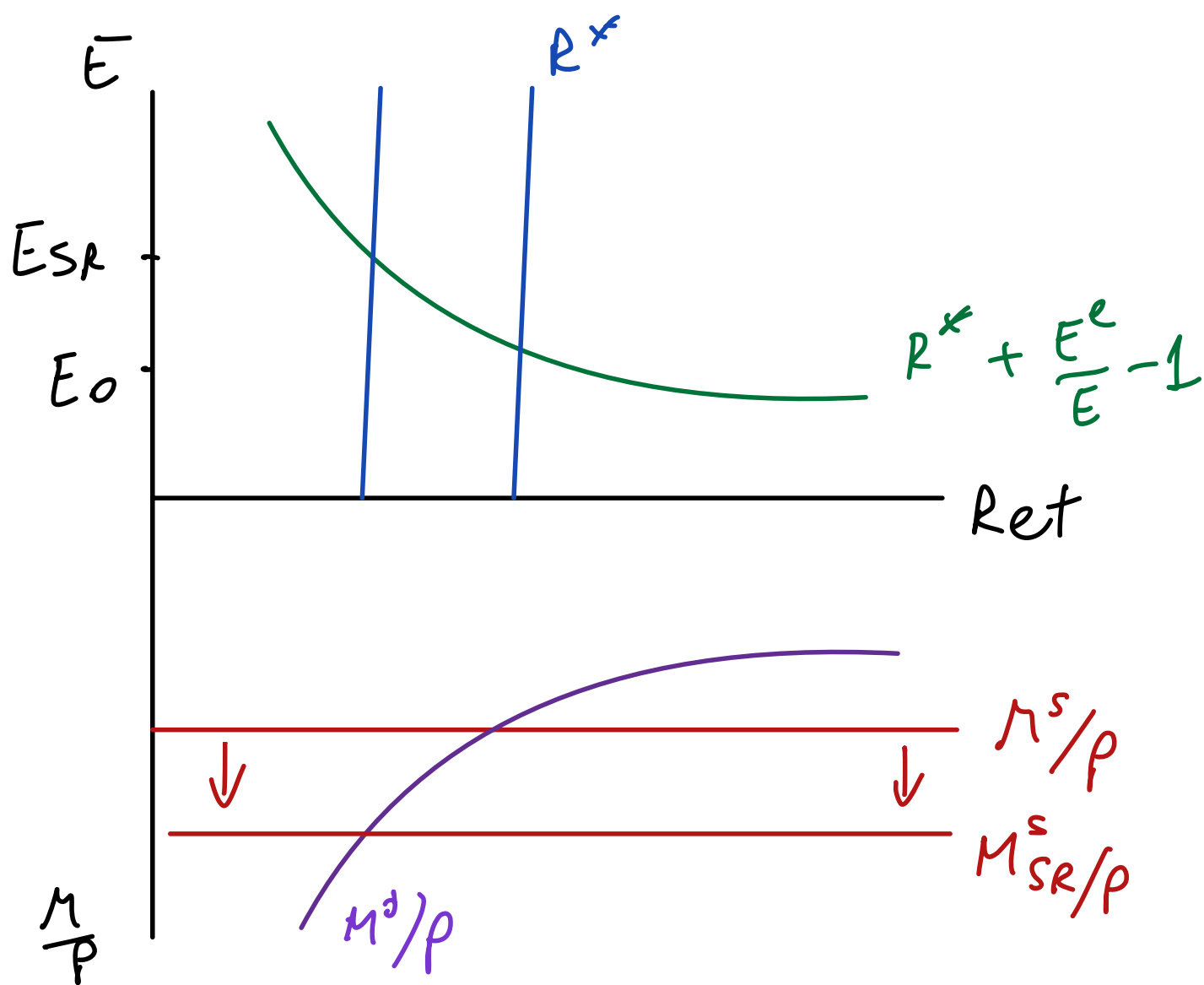
- KEEP LONG RUN E

UNCHANGED $\Rightarrow$

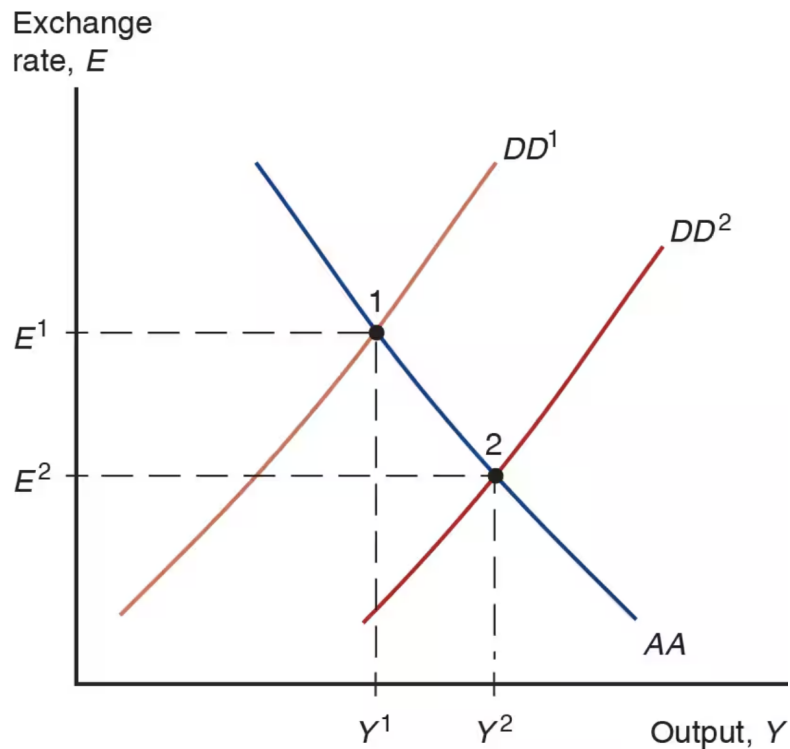
$E_{sr}^e = E_{LR}$ UNCHANGED

- $P_{sr} = P_0$ FIXED

$M^S \uparrow$
 LEADS TO
 $E \uparrow$



Temporary Change in Fiscal Policy



$G \uparrow$ IN SR ONLY

$D \uparrow$ $Y \uparrow$ FOR ANY E

$\Rightarrow$ DD SHIFTS TO
THE RIGHT

Example: DD Schedule

$$C_{(+)}(Y_D) = 1 + 0.75Y_D$$

$$EX_{(+)}(q, Y^*)_{(+)} = 0.15 + 0.5q + 0.1Y^*$$

$$IM_{(-)}(q, Y_D)_{(+)} = 0.1 - 0.2q + 0.12Y_D$$

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

- Plug in given C , EX , IM , Y_D , q into the equilibrium condition for the goods market $Y = C + I + G + EX - IM$ to get:

$$Y = 1.05 + 0.63(Y - T) + 0.7 \frac{EP^*}{P} + I + G + 0.1Y^*$$

- Solve for E to get the DD curve:

$$E = \frac{1}{0.7} \frac{P}{P^*} (-1.05 + 0.37Y + 0.63T - I - G - 0.1Y^*)$$

Example: AA Schedule

$$R = R^* + \frac{E^e}{E} - 1$$

$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\frac{M^d}{P} = L(\underset{(-)}{R}, \underset{(+)}{Y}) = 1.1 - R + 0.05Y$$

- Plug in UIP into money market equilibrium condition:

$$\frac{M^s}{P} = 1.1 - \left(R^* + \frac{E^e}{E} - 1 \right) + 0.05Y$$

- Solve for E to get the AA curve:

$$E = \frac{E^e}{2.1 - \frac{M^s}{P} - R^* + 0.05Y}$$

Full AA-DD Model

DD Schedule: $Y = C(Y - T) + I + G + CA(EP^*/P, Y - T, Y^*)$

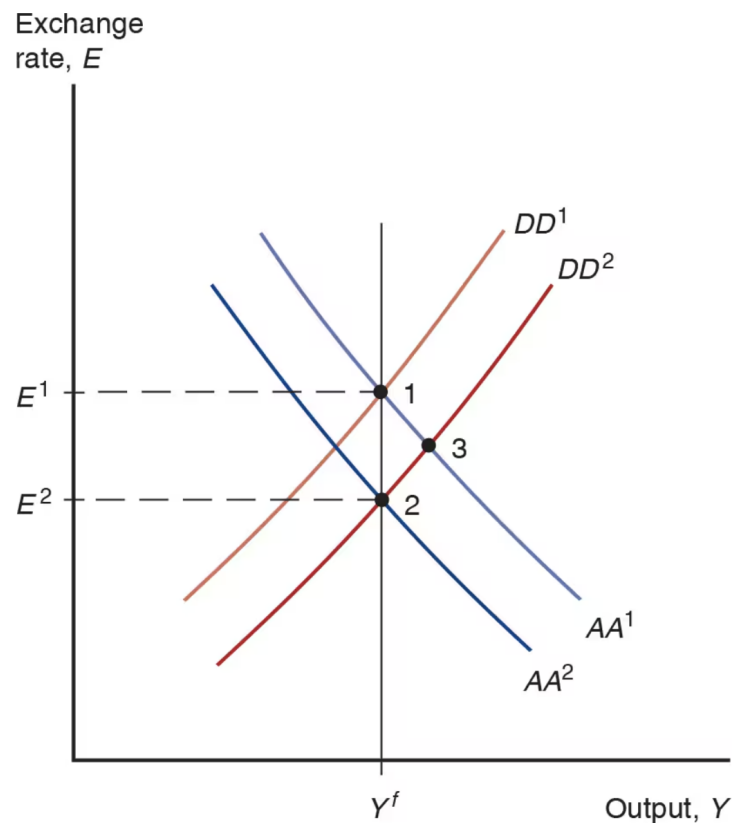
AA Schedule: $\frac{M^s}{P} = L\left(R^* + \frac{E^e}{E} - 1, Y\right)$

Phillips Curve: $\pi = \pi^e + \alpha(Y - Y^f)$

Definition of inflation: $\pi_t = \frac{P_t}{P_{t-1}} - 1$

Definition of expected inflation: $\pi_t^e = \frac{P_{t+1}^e}{P_t} - 1$

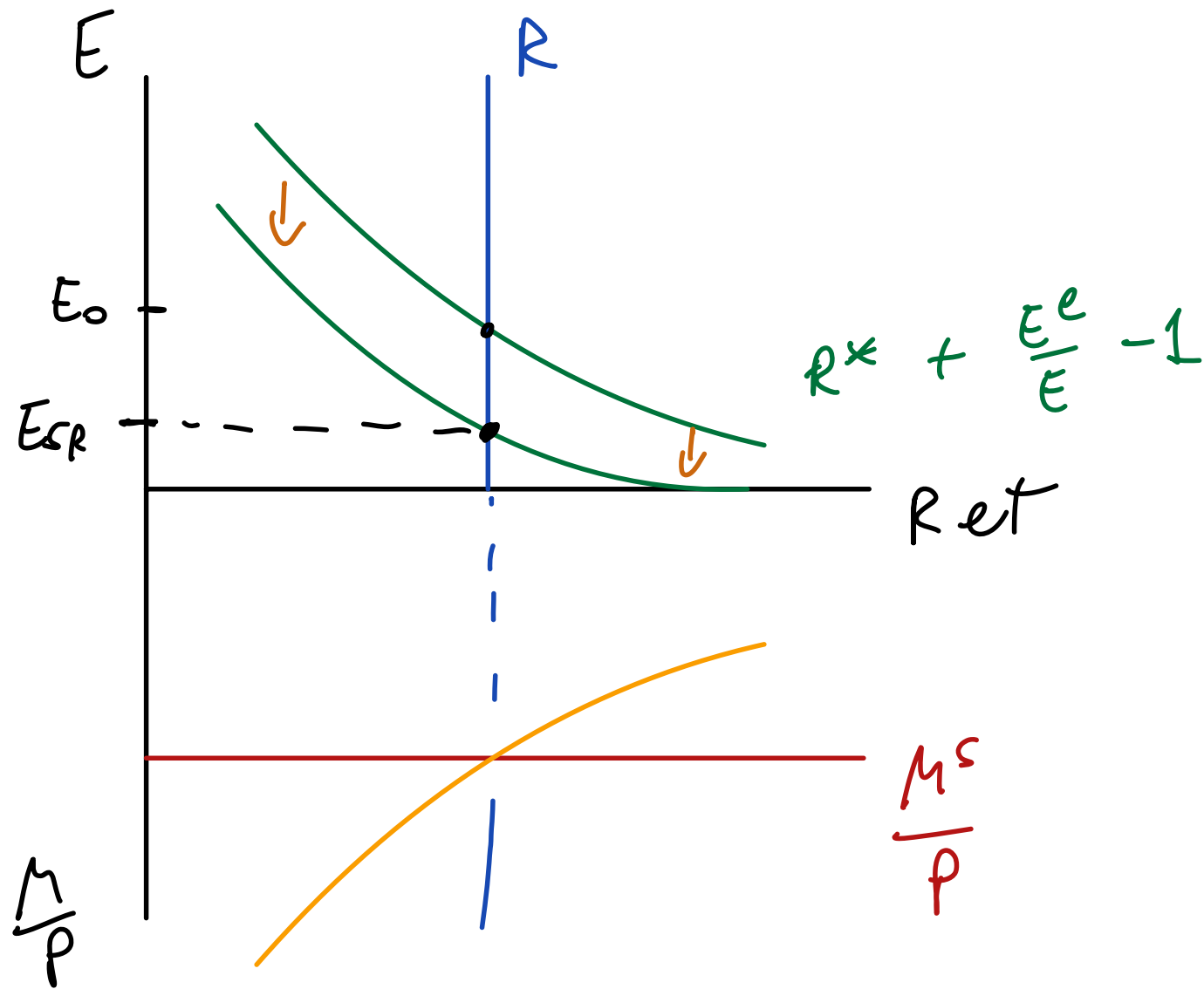
Permanent Shifts in Fiscal Policy



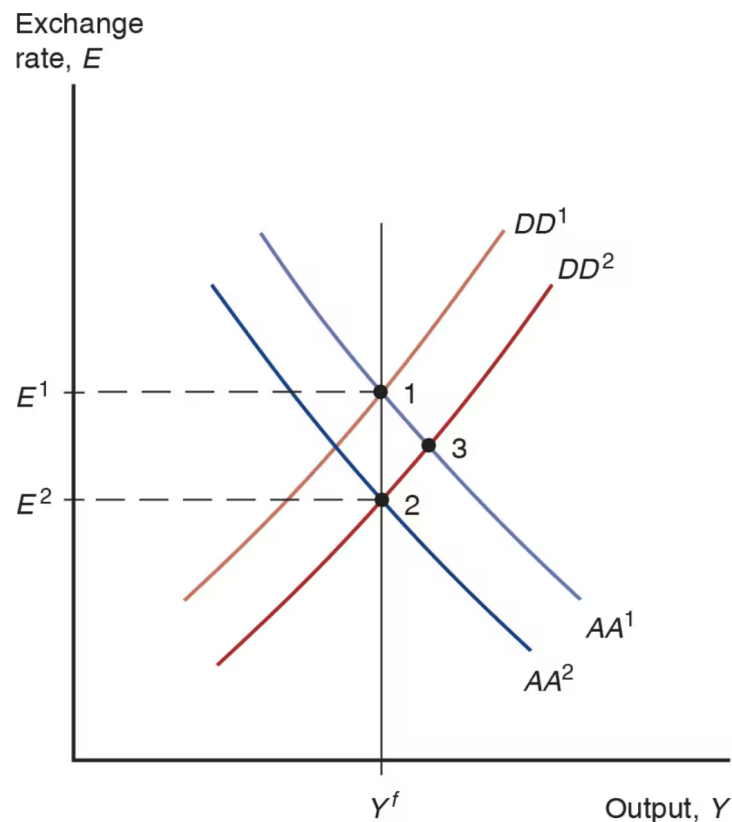
$G \uparrow$ PERMANENTLY

$\Rightarrow D \uparrow \Rightarrow Y \uparrow \Rightarrow DD$ SHIFTS
TO THE
RIGHT

$$(AA) : \frac{M^s}{P} = L \left(R^* + \frac{E^e - E}{E}, Y \right)$$

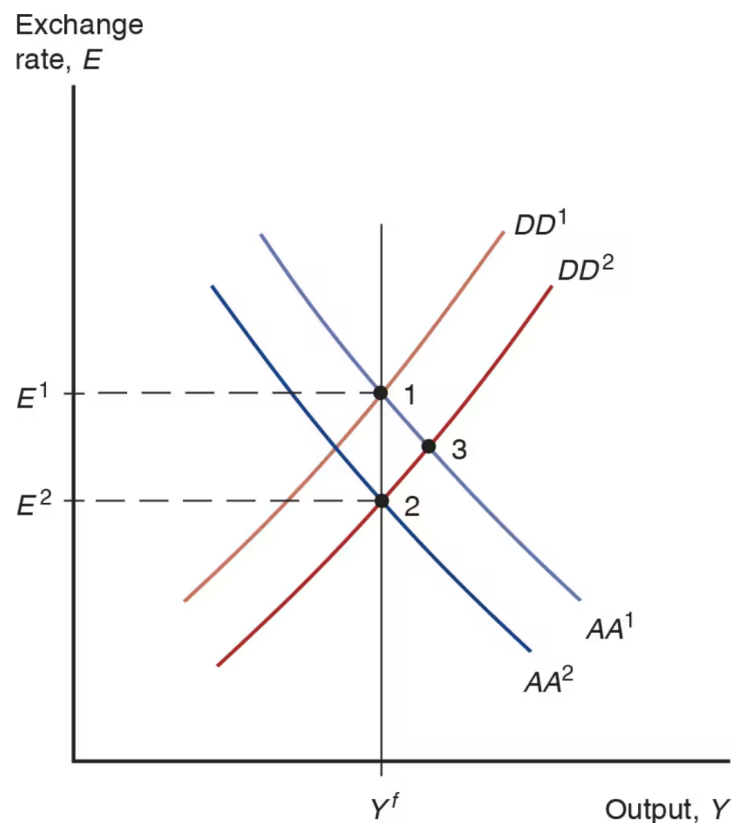


Permanent Shifts in Fiscal Policy



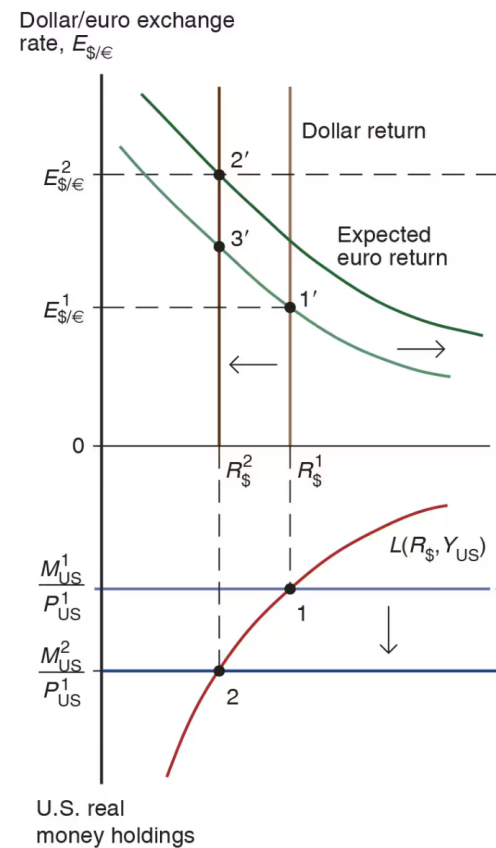
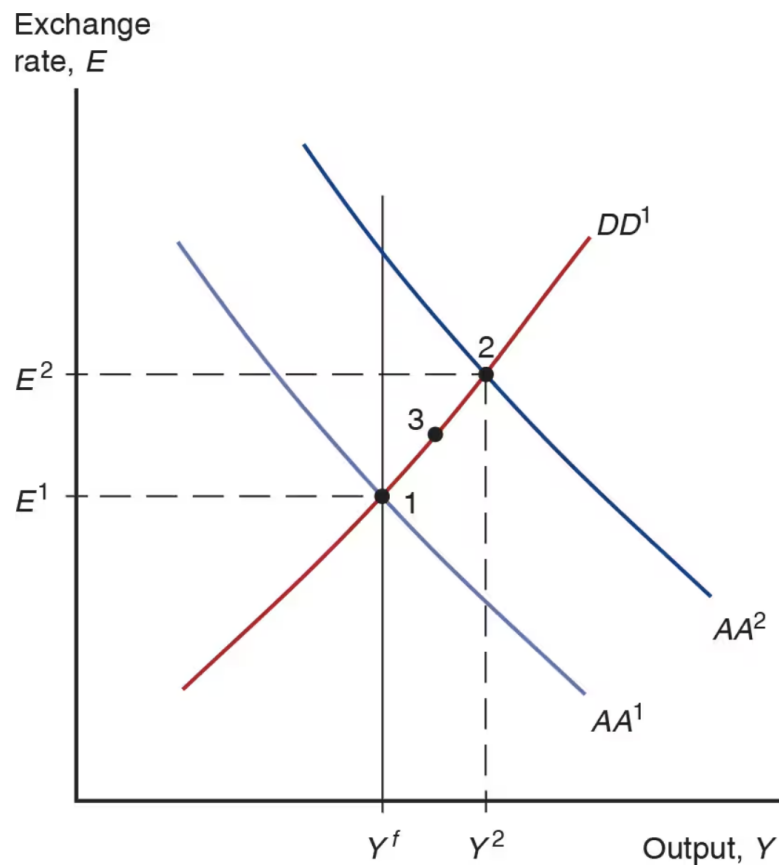
- Start in a medium run equilibrium at point 1 with $R_0 = R^*$ and $E_0^e = E^1$
- Increase in G shifts DD to the right
- At point 3, exchange rate is lower than E_1
- Since the shift in DD is permanent, appreciation at point 3 is also expected to be permanent
- E^e must also go down
- Lower E^e shifts AA down
- But how much?

Permanent Shifts in Fiscal Policy



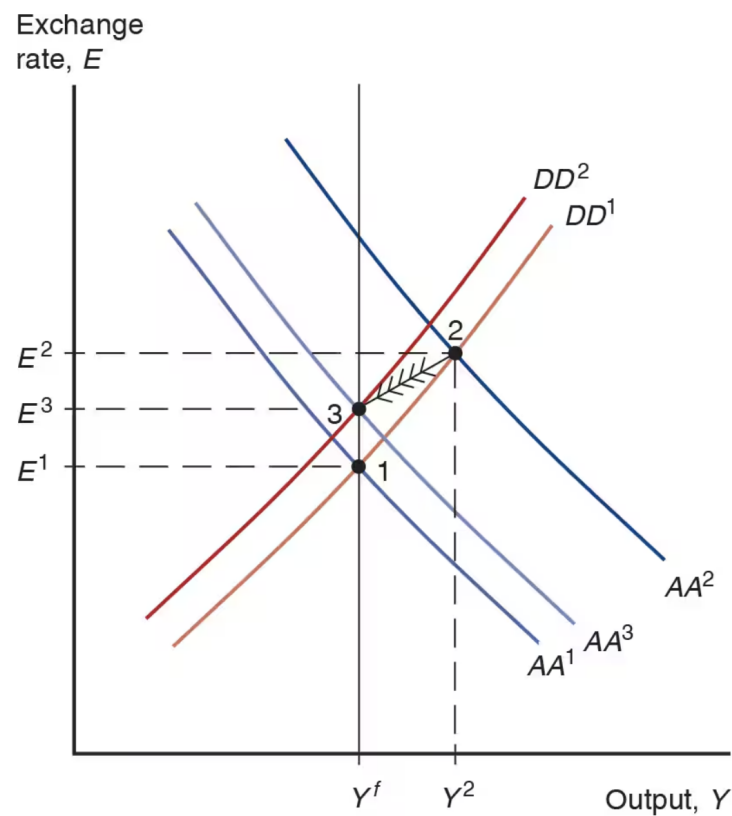
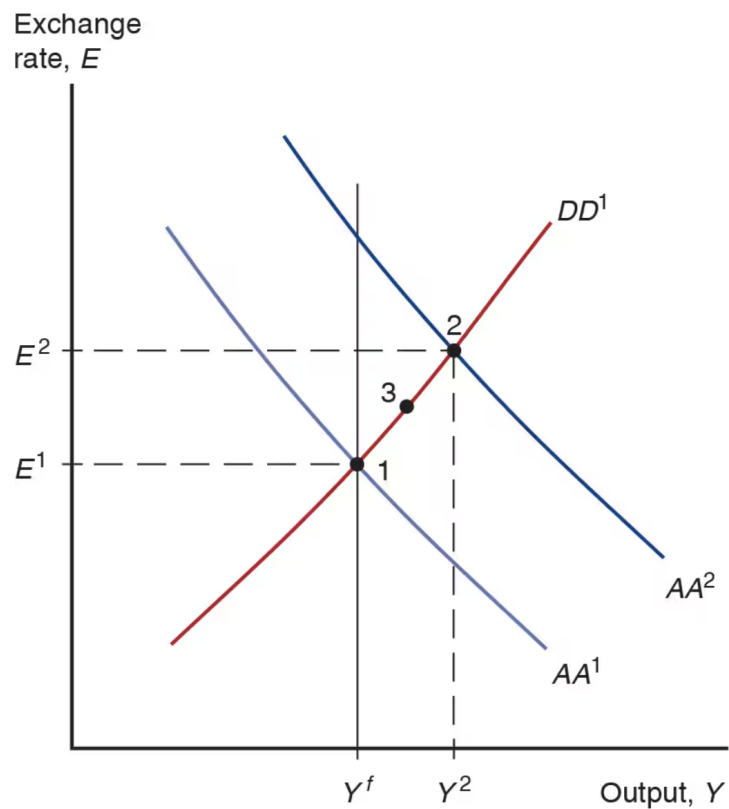
- There are no more shifts expected to occur, E^e remains constant
- Real money supply is constant
- Long run real money demand must stay unchanged
- Then P never changes
- $MS=MD$ implies R does not change
- UIP implies E does not change either
- If Y must be equal to its original level Y^f , AA^1 must shift to AA^2

Permanent Shifts in Monetary Policy

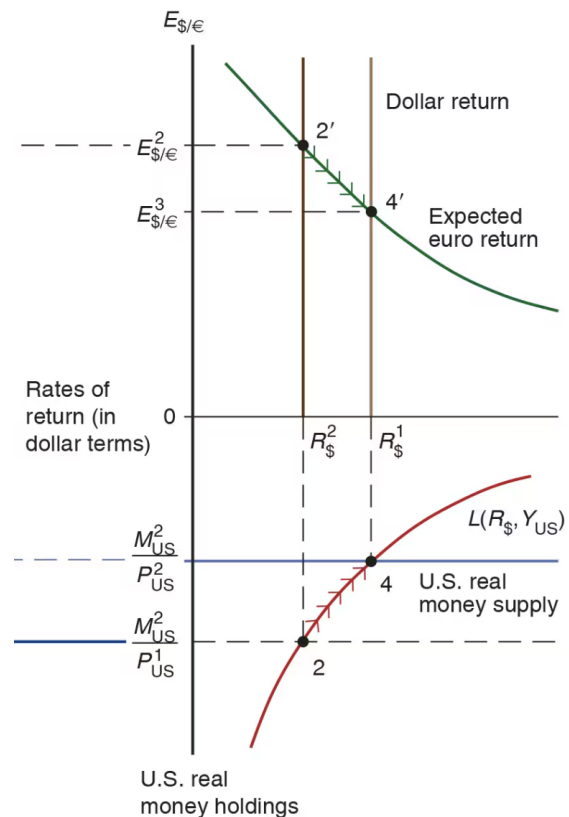
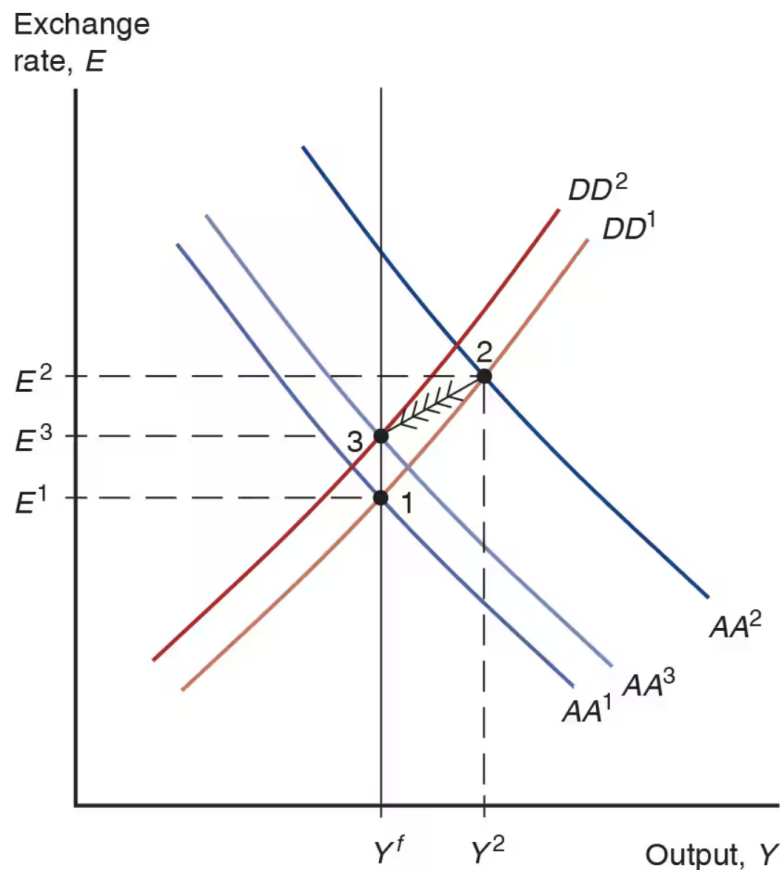


(a) Short-run effects

Permanent Shifts in Monetary Policy



Permanent Shifts in Monetary Policy



(b) Adjustment to long-run equilibrium

DD Schedule with Tariffs

IMPORT TARIFFS - "AD VALOREM" $\rightarrow$ PROPORTIONAL

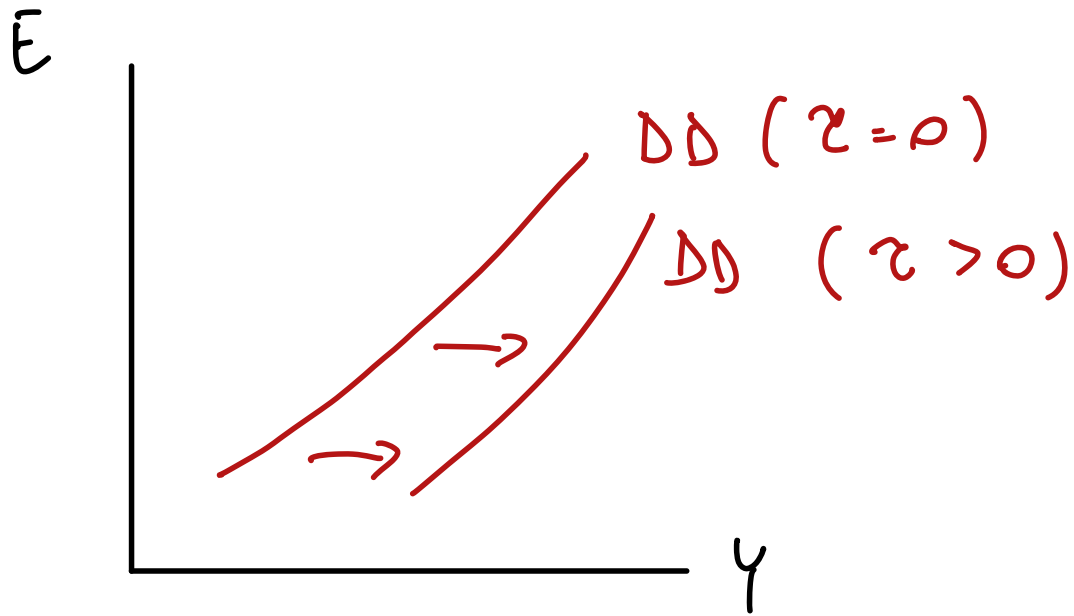
$$\begin{array}{l|l} \underbrace{IM}_{\substack{\text{UNITS} \\ \text{OF DOM} \\ \text{GOOD}}} = q \times \underbrace{VOLUME}_{\substack{\text{IN UNITS} \\ \text{OF FOREIGN} \\ \text{GOOD}}} & IM(q, Y-T) \\ & \begin{array}{l} \downarrow \\ \text{CHANGE} \\ \text{OF} \\ \text{UNITS} \end{array} & \begin{array}{l} (-) \quad (+) \\ q \times VOLUME(q, Y-T) \\ (-) \quad (+) \\ q \times VOLUME\left(\frac{EP^*}{P}, Y-T\right) \end{array} \end{array}$$

$$IM = q \times VOLUME \left(\frac{EP^x}{P} (1 + \tau) , Y - T \right)$$

AD VALOREM TARIFF

$$IM = IM \left(\underset{(-)}{q} , \underset{(+)}{Y - T} , \underset{(-)}{\tau} \right)$$

$$Y = C + I + G + EX - IM \left(\underset{(-)}{\tau} \right)$$



DD SHIFTS TO
THE RIGHT
WHEN z
GOES UP